Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Instructions

1. Which of the following statement(s) is/are true about a binary logistic regression? 1 / 1 point

(I) The predictor variables (x) are continuous, with values between 0 and

(II) The response variable (y) is continuous, with values between 0 and 1.

(III) The outputs of the sigmoid function are continuous, with values between 0 and 1.

- ○ I only
- ◉ II only
- ◉ III only
- ○ II and III only
- ○ I, II, and III

Correct

2. Let Y denote whether a student get admitted to attend Illinois Tech, with $Y = 1$ as successfully being admitted, and $Y = 0$, otherwise. Also, let 1 / 1 point

- X_1 denote the math scores (0-100)
- X_2 denote the reading scores (0-100)

of a particular standardize exam that a student need to take to gain admission. We collected n-data and run a logistic regression to obtain

$$\ln\left(\frac{\pi}{1-\pi}\right) = 0.13 + 0.05X_1 + 0.04X_2$$

where

$$\pi = \Pr(Y = 1|X_1 = x_1, X_2 = x_2).$$

Suppose a student has a math scores of 78% and the reading scores of 82%. If the student retake the math test and scores 83%. Compute the following change of odds in being admitted to Illinois Tech.

$$\frac{\text{Odds}(x_0+1) - \text{Odds}(x_0)}{\text{Odds}(x_0)} \times 100\%$$

- ◉ 5.127%
- ○ 13.883%
- ○ 4.081%
- ○ 0.05%
- ○ 0.04%

Correct

3. Suppose we want to model the profitability of a movie (y) based on the opening-weekend revenue (x, in millions of dollar). We let $y = 1$ if the movie is profitable, that is, the revenue is larger than the budget, and not profitable as $y = 0$. Based on the data from 100 movies in the past 50 years, we obtain 1 / 1 point

$$\log(\text{odds}) = -2.27 + 1.276x$$

Which of the following is/are correct?

(I) The odds ratio is given by $e^{1.276}$.

(II) For a one-unit increase in x, we expect a 258.2% increase in the odds.

(III) For a one-unit increase in x, we expect the odds that the movie will be profitable increase by a factor of 3.58.

- ○ I only
- ○ II only
- ○ III only
- ○ II and III only
- ◉ I, II, and III

Correct

4. Suppose that the probability of an event is 0.2, what is the odds of the event occurring? 1 / 1 point

- ○ 0.2
- ○ 0.8
- ○ 0.12
- ◉ 0.25
- ○ 0.3

Correct